

Survival analysis and state transition models

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A broad definition of *survival analysis* would include any research setting in which we are monitoring units that can transition between discrete states over time. The research aim of the survival analysis could be to gain an understanding any aspect of these transitions, such as the rate at which transitions occur, the distribution of dwell times in each state, statistical dependence between transitions, and how any of these phenomena are related to covariates.

Most commonly, survival analysis methods are used with *time to event* or *duration* data, where “time to event” refers to the duration of time from an origin until some event of interest occurs, which can be viewed as a state transition. In the most basic example, there are usually only two states, typically described as “alive” and “dead”. All subjects begin in the “alive” state and eventually transition to the “dead” state, which is “absorbing”.

A typical example with more than two states would be one in which subjects can be “healthy”, “ill”, or “dead”, with all transitions allowed except that the “dead” state is absorbing – once reached, there is no possibility to transition to another state.

An important aspect of survival analysis is that there is often only partial information about each subject’s status, usually due to incomplete monitoring. For example, in a health study we would rarely be able to wait until all subjects have died, so some subjects will end the study in the “alive” state. This is an example of *censoring*, which will be defined more formally below.

1. Key concepts

1.1. Time origin

Consider a setting where we are monitoring a person over time, and are interested in the time T at which some event of interest occurs. To begin, it is important to explicitly define the time units^o, e.g. days, months, or years, and the *origin* from which time is measured, i.e. what is the meaning of time $T = 0$? For example, if T denotes the age of a person when an event of interest occurs, the time origin is the date of birth. Alternatively, there may be an event that must occur before the event of interest, and the time of this event may make a more sensible time origin. For example, if T corresponds to graduating from university, we may choose the date of university matriculation (first enrollment) as the time origin, so that e.g. if $T = 4$ and the time units are years, then the person graduated four years after beginning their studies.

1.2. Event time distributions

The theoretical basis of conventional survival analysis is that we are studying the probability distribution of a *random variable*^o T , corresponding to the time at which an event of interest occurs. In conventional survival analysis, the event will always occur if we wait long enough, so $P(T < \infty) = 1$. In some situations this may not be realistic and there is a subdomain of survival analysis called *cure modeling* in which $P(T < \infty) < 1$ is allowed. However here we deal exclusively with the conventional setting where $P(T < \infty) = 1$ is assumed.

1.3. Censoring

Data in a survival analysis are often subject to “censoring”, which means that we only have partial information about the value of T for many subjects. Suppose we are analyzing data from a medical study where we are studying the incidence of a complication following a

medical procedure. For example, if the population consists of people who have had a cardiac stent implanted, then a common complication would be stent thrombosis (a clot forming near the stent). In this case, we may use age in years as our time scale and let T denote the age when a subject first has stent thrombosis.

Only a subset of the subjects will have stent thrombosis during our study. Other subjects will be followed for a period of time and will never be observed to have stent thrombosis. Let R denote the last age at which the person could be observed. If $T < R$, we observe T but if $T > R$ we do not know the value of T . More formally, we observe the time $Y = \min(T, R)$ and the *status indicator* $\delta = \mathbf{I}(Y = T)$. This is called **right censoring**^o – we know that the value of T is greater than some known value, but we do not know the exact value of T for many subjects. In this context, R is known as the *right censoring time*.

In some settings it is important to emphasize that R is the last time at which the subject's state could be observed, irrespective of whether they have already had the event by that time. Whether we know the exact value of R may depend on the nature of the censoring. If a subject is lost to follow up and $R > T$, we will not know the exact value of R , but if R represents a subject's age at the planned end of the study (after which data collection will cease), we may have $R > T$ but still know the exact value of R (assuming that the subject cannot drop out before the end of the study).

Right censoring is the most commonly encountered form of censoring, but in some settings we may have *left censoring*, meaning that we only know that T is less than some observed value. Also, there is *interval censoring* in which we know, for example, that someone had stent thrombosis between the age of 75 and 77 but we do not know the exact age at which the stent thrombosis occurred.

A common assumption in survival analysis is that of *independent censoring*. We will define this here in the context of right censoring. As above, let T be the event time and R be the right censoring time. We usually cannot observe both T and R . Nevertheless, we can imagine that both values exist, one being a “latent” value. Independent censoring simply means that T and R are independent random variables (in a regression analysis, we may require the weaker condition that T and R are independent given the covariates). This independence implies, for example, that people who are prone to having an early event (e.g. unhealthy people) do not have systematically different censoring times than people who typically have late events.

Since we don't observe T and R together, independent censoring is almost always an untestable assumption. In some cases, based on the study design or other external information, there may be reason to accept it and in other cases there may be good reason to doubt that independent censoring holds. There are various methods for effectively handling dependently censored data, but that is an advanced topic that we will not consider further here.

1.4. Truncation

An important concept in survival analysis is the potential selection bias induced by *truncation* or *delayed entry*. If units (e.g. people) are selected into the sample conditionally on their event time, then this must be taken into account. The most common form of truncation is *left truncation* in which there is a value L (which may be specific to each observation) such that if

the event occurs before time L (i.e. if $T < L$) then the person would not have been included in our sample.

For example, suppose in our analysis of stent thrombosis following stent placement that we only have access to data from one health insurance provider. People may join this health insurance program at any age, and we can let L denote the age at enrollment. If a person had a stent placed before joining this insurance company's plan, we would not know the exact age when the stent was placed. Therefore, we may choose to eliminate from the analysis all people who already had a stent when joining the health insurance plan. Not having a stent at the beginning of the insurance record is a requirement for being in our study, so the age at which a person began their health insurance coverage would be a left truncation time.

More formally, if we have left truncation then we are working with the conditional distribution $P(T \mid T \geq L)$, while if there is no left truncation then we are working with the unconditional distribution $P(T)$. If we are doing regression analysis with covariates X , then with left truncation we are studying $P(T \mid T \geq L, X)$ and with no such truncation we are studying $P(T \mid X)$. If the research aim is based on $P(T \mid X)$ but we have left truncation, we usually will want to adjust the results to account for the difference between $P(T \mid T \geq L, X)$ and $P(T \mid X)$.

Just as censoring can be independent or informative, left truncation also has this property. The left truncation is said to be independent if L is independent of (T, C) , which in practice is assessed by comparing L to Y . Independent left truncation can be easily handled with “risk set adjustment” methods to be discussed below, but informative left truncation requires more care. Typically an informative left truncation process is modeled, and inverse probability of inclusion weights are used to account for it.

1.5. Competing risks

In a survival analysis there may be other events that “compete” with the event of interest. For example, if we are studying the time T at which a person has a stroke, it is possible that the person dies of another cause before having a stroke. Death unrelated to stroke is a *competing risk* for the event of interest. This can be viewed as another example of a multi-state transition model, with states “healthy”, “had stroke”, and “dead”. Each person has a stroke age T , a death age D , and a censoring age R . Conceptually, there is a joint distribution for (R, D, T) , but if $D < T$ we will not know the value of T . People can transition from “healthy” to “had stroke”, from “had stroke” to “dead”, and from “healthy” to “dead”. The remaining transitions have zero probability.

Superficially, D appears to be another form of censoring, so we may consider redefining R as $R' = \min(R, D)$. But doing this may create dependent censoring. Even if R is independent of T , R' may not be, for example if less healthy subjects are at greater risk for both stroke and death.

1.6. The risk set

The *risk set* at a specific time t is the subset of units (e.g. people) who could possibly experience the event at time t . Anyone who has already had the event before time t , has been right

censored before time t , has experienced a competing risk before time t , or is truncated after time t is not in the risk set at time t .

1.7. Recurrent events

In conventional survival analysis, each subject experiences the event exactly once (although we may not observe this occurrence in our study due to censoring). However some events may be able to recur. For example, suppose that we are studying arrests by police and the time T is the duration until a person is arrested. There can be subsequent arrests for the same individual and it may be of interest to study the distribution of event times for all “spells” between one arrest and the next arrest.

This setting can be viewed as another example of a multi-state model. In the example, we would have a state for “never arrested”, “arrested 1 time”, “arrested 2 times”, etc. Subjects can transition from each state to the next one in sequence.

1.8. Counting process notation

Subjects who are right-censored have a maximal *follow-up time* which is the greatest time at which they were observed and confirmed not to have yet experienced the event of interest. In survival analysis with right censoring, where T_i is the (possibly unobserved) event time and R_i is the time at which we would no longer be able to observe the subject, we typically write $Y_i = \min(T_i, R_i)$ as the “observed” time, which is either the follow-up time for censored subjects or the event time for non-censored subjects. Then, we define the *status indicator* δ_i such that $\delta_i = 1$ if the event is observed and $\delta_i = 0$ if the event is not observed. Note that when $\delta_i = 1$, then $Y_i = T_i$, and if $\delta_i = 0$ then $T_i > Y_i$.

A more general notation that is often encountered is that for each subject we have an interval $[L_i, R_i)$ such that the event is known to occur within this interval. For right censored subjects, $R_i = \infty$. For non-censored subjects, $L_i = R_i$. An *interval censored* subject has $0 < L_i < R_i < \infty$, and a left-censored subject has $0 = L_i < R_i < \infty$. For more general recurrent event data or when there are more than two states, we can decompose each individual’s history into a series of disjoint records L_j, Y_j, δ_j , where L_j denotes the time at the beginning of the record, Y_j denotes the time at the end of the record, and δ_j denotes the state at the end of the record. Here, instead of δ_j being a binary variable, it is an indicator of the state occupied by the subject at the end of the interval.

In counting process notation, for the setting with two states (alive/dead with dead being absorbing), we have a non-decreasing function of time E_i such that $E_{i(t)} \in \{0, 1\}$ for each t . This function reflects whether unit i has died on or before time t . The derivative $dE_i(t)$ is equal to 1 at the time t where death occurs, and is equal to zero everywhere else. It is not a derivative in the usual sense, but formally acts as such. For example, $E_i(t) = \int_{s=0}^t dE_i(s)ds$.

[Here](#)[◦] and [here](#)[◦] are two thorough treatments of the use of counting process notation in survival analysis.

2. Estimation of the survival function

The *survival function* of a random variable T is defined to be $S(t) \equiv P(T > t)$. It is closely related to the [cumulative distribution function](#)^o (CDF), defined to be $F(t) = P(T \leq t)$ since $S(t) = 1 - F(t)$. In words, the survival function at time t is the probability that the event has not occurred by time t . Another name for the survival function is the *complementary CDF*. We may also refer to this as the *marginal survival function* to emphasize that it is not conditioned on any covariates.

The empirical CDF (eCDF) is one of the fundamental objects in statistics. Based on an independent and identically distributed (IID) sample from some distribution, the eCDF is defined as $\hat{F}(t) = \#\{T_i \leq t\}/n$. If there is no truncation and if the values of T_i are all observed (i.e. there is no censoring) then we can estimate the survival function as $\hat{S}(t) = 1 - \hat{F}(t)$.

In survival analysis we usually have censoring and/or truncation. We will consider here only the important subcase where there is right censoring and no truncation. In this setting there is a simple estimator of the survival function $S(t)$ known as the *product limit* estimator or the [Kaplan-Meier](#)^o estimator.

The Kaplan-Meier estimator focuses exclusively on the observed event times. Let $t_1 < t_2 < \dots < t_m$ denote the distinct times at which events are observed to occur, let e_i denote the number of events that occur at time t_i , and let n_i denote the size of the risk set just before time t_i . The estimated probability of passing through time t_i without experiencing the event is $1 - e_i/n_i$. Thus, the estimated probability of making it from time 0 to time t without experiencing the event is

$$\hat{S}(t) \equiv \prod_{i:t_i \leq t} \left(1 - \frac{e_i}{n_i}\right). \quad (1)$$

This is the product limit estimator of the survival function. Note that the actual survival function can be any non-increasing right continuous function, and thus in general $S(t)$ will change at infinitely many values of t . However the Kaplan-Meier estimate of the survival function is a step function that only changes at the observed values of t where an event occurs (just as the eCDF only changes at the observed data values). A basic exercise is to prove that in the absence of censoring, the product limit estimate $\hat{S}$ is the complement of the eCDF, i.e. $\hat{S} = 1 - \hat{F}$.

While the primary object of interest in survival analysis is the probability distribution of event times T in the absence of censoring, there are also situations where the censoring distribution itself is of interest. For example, we may want to document that the subjects in a study are followed for a sufficiently long time to achieve high power at all relevant time points. A useful application of the Kaplan-Meier estimate is the so-called *reverse Kaplan-Meier* plot, which estimates the censoring distribution $P(R)$. This is obtained by swapping δ so that $\delta' = 1 - \delta$. The resulting probabilities estimate the survival function for the censoring process. This approach assumes independent censoring.

2.1. Left truncation

The survival function can be estimated in the presence of left truncation using *risk set adjustment*. This simply means that the risk set at time t_j excludes units whose left truncation time L is greater than t_j . Specifically, the risk set at time t is

$$\text{Risk}(t) = \{i : L_i \leq t \leq Y_i\}. \quad (2)$$

This modified version of the Kaplan-Meier estimator is sometimes known as the *Lynden-Bell estimator*.

The estimation target of this estimator depends on the support of the left truncation distribution $P(L)$. If this support has a lower bound L^* , then the estimation target is the conditional survival function given that the survival time is at least L^* , $P(T > t \mid T \geq L^*)$. A special case is “administrative left truncation” in which everyone has the same left truncation time (say everyone enters a study at age 18). In this example, the Kaplan-Meier/Lynden-Bell estimate converges to $P(T > t \mid T \geq 18)$.

Risk set adjustment requires the left truncation to be independent of the event and censoring times, otherwise the resulting estimate of the survival function may be severely biased. This independence can be assessed using *Tsai’s test*, which is essentially Kendall’s tau correlation adapted to handle truncation and censoring. Let $Y_i = \min(T_i, R_i)$ be the observed event or censoring time (the status indicator δ_i does not play a role here). We will only consider pairs of observations i, j that are “comparable”, in the sense that the indicator

$$Q_{ij} = I(\max(L_i, L_j) < \min(Y_i, Y_j)) \quad (3)$$

is equal to 1. This is necessary so that if we swap Y_i and Y_j both observations would remain non-truncated. The test statistic is

$$\frac{\sum_{i < j} Q_{ij} \cdot \text{sgn}(L_i - L_j) \cdot \text{sgn}(Y_i - Y_j)}{\sum_{i < j} Q_{ij}}. \quad (4)$$

The statistic can be calibrated using an asymptotic variance approximation or using permutation inference (randomly swapping Y_i and Y_j within each pair with probability 1/2).

2.2. Inference for the survival function

There are many methods for statistical inference relating to survival functions. The [log rank test](#)[◦] is a formal [hypothesis test](#)[◦] of the null hypothesis that two survival functions are equal, i.e. the null hypothesis $S_0(t) \equiv S_1(t)$. It is also possible to put both pointwise and simultaneous confidence intervals (bands) around the estimated survival function $\hat{S}(t)$ to convey the precision with which it is estimated.

Here we briefly derive *Greenwood’s formula*, which is a simple method for obtaining approximate standard errors for $\hat{S}(t)$ at a fixed value of t . Let $\hat{p}_i = 1 - e_i/n_i$, so that $\hat{S}(t) = \prod_{i:t_i \leq t} \hat{p}_i$. We work on the log scale

$$\log(\hat{S}(t)) = \sum_{i:t_i \leq t} \log(\hat{p}_i). \quad (5)$$

By linearization,

$$\text{var}[\log(\hat{p}_i)] \approx \hat{p}_i^{-2} \hat{p}_i(1 - \hat{p}_i)/n_i = (1 - \hat{p}_i)/(n_i \hat{p}_i) = e_i/(n_i(n_i - e_i)). \quad (6)$$

Thus

$$\text{var}[\log(\hat{S}(t))] \approx \sum_{i:t_i \leq t} \frac{e_i}{n_i(n_i - e_i)}. \quad (7)$$

Another application of linearization allows us to remove the logarithm, yielding

$$\text{var}[\hat{S}(t)] \approx \hat{S}(t)^2 \sum_{i:t_i \leq t} \frac{e_i}{n_i(n_i - e_i)}. \quad (8)$$

3. Hazard functions

The [hazard function](#)^o is a way of mathematically representing a probability distribution that is commonly used in survival analysis. The hazard function is defined to be

$$h(t) \equiv \lim_{\delta \downarrow 0} \frac{P(T \leq t + \delta \mid T \geq t)}{\delta} = \lim_{\delta \downarrow 0} \frac{S(t) - S(t + \delta)}{\delta S(t)}. \quad (9)$$

For simplicity this definition does not consider left truncation, but the generalization is simple. The hazard function can be interpreted as the “instantaneous event rate”. It has units of 1/time so is not dimensionless (the time units matter).

It takes some practice to understand how to interpret this limit. If the time unit is “days” and the hazard is 0.001 at day 100, then this means that approximately 0.1% of the subjects at risk on day 100 will experience the event on that day. Note that this is an approximate statement since we are not actually taking a limit here. This approximate statement is closer to being true over time intervals where the hazard function is approximately constant.

Note that the hazard function is not a probability and can be greater than 1 (but it must be non-negative). It is also not a density, although if the density exists it can be determined from the hazard function. A hazard greater than 1 can be interpreted as follows – if subjects were able to be resurrected and rejoin the risk set immediately after experiencing the event, then the number of events in a short time interval could be greater than the number at risk at the beginning of the interval.

When the survival function is smooth, the hazard function is also the logarithmic derivative of the survival function:

$$h(t) = -\frac{d}{dt} \log S(t) = -\frac{S'(t)}{S(t)}. \quad (10)$$

Another important quantity to understand is the *cumulative hazard function*

$$H(t) = \int_0^t h(s) ds. \quad (11)$$

The identity $S(t) = \exp(-H(t))$ holds for the following intuitive reason. First, take a sequence $t_1 < t_2 < \dots$, and approximate $S(t)$ using a product

$$S(t) \approx \prod_{i:t_i \leq t} (1 - h(t_i)(t_{i+1} - t_i)). \quad (12)$$

This is the same idea as used in the Kaplan-Meier estimator, but instead of working with data, here we are approximating the population survival function. Next use the approximation $1 - h(t_i)(t_{i+1} - t_i) \approx \exp(-h(t_i)(t_{i+1} - t_i))$. This allows us to convert the product into a sum

$$S(t) \approx \exp\left(\sum_{i:t_i \leq t} -h(t_i)(t_{i+1} - t_i)\right) \approx \exp(-H(t)). \quad (13)$$

If T has a density f , then $f = F' = -S'$. Thus

$$f(t) = h(t) \exp(-H(t)). \quad (14)$$

Note that this also implies that when densities exist, $h(t) = f(t)/S(t)$, giving another natural view of the hazard function.

Finally, we can introduce the notion of a *product integral* and find that

$$S(t) = \prod_0^t (1 - h(s)) ds. \quad (15)$$

Above, $\prod$ represents a product integral, which is a limit of products over finite partitions of $[0, t]$, just as the usual integral is a limit of sums over finite partitions of $[0, t]$. The factors in the product have the form $1 - h(s)\Delta s$, where Δs is the width of a small interval containing s .

In many applications, the hazard function may be easier to interpret than the survival function. A common consideration is whether the hazard function is increasing, decreasing, approximately constant, or has some other shape like a “U” (“bathtub”) shape. In terms of parametric distributions, the exponential distribution has a constant hazard function, whereas the Weibull distribution can have either an increasing or decreasing hazard function depending on its parameters.

One common application of survival analysis is in the setting of failures of manufactured products, e.g. how likely is it that your car will break down at a particular point in time, given that it is currently operational? If the hazard function is constant, then the car is equally likely to break down on every day that you own it. If the hazard function is increasing, then as the car gets older it becomes more likely to break down. This could be due to the parts of the car wearing out and failing with use (e.g. due to material fatigue).

If the hazard function is decreasing then as the car gets older it becomes less likely to break down. This could occur if most failures are due to manufacturing flaws. A car consists of thousands of parts, and suppose that in any given car, a few parts may be flawed at the

point of manufacture, or are installed incorrectly during manufacture. These flaws may not be sufficient to cause the car to break down immediately, hence the car is sold in apparent working order. However these flawed parts may tend to fail very early. Many cars need to be serviced early in their lifespans for this reason. However, once you have driven your car say 5000 miles, these flaws have been discovered and resolved and the risk of a break-down becomes lower with time (while the other type of failure due to wear out and fatigue has not yet become common).

With manufactured products, both of the above mechanisms are likely to exist, so the hazard function may exhibit a “bathtub” shape. This means that the hazard function is higher for small t due to manufacturing errors, then the hazard function is lower once the car has been driven long enough to identify and resolve any such issues. But as the car gets older and parts fatigue, a different type of failure becomes more likely and the hazard function begins to increase again.

A similar phenomenon exists with human lifespans, whereby the hazard of dying is greater for infants and very young children (up to around age 3-5) and then becomes very low for several decades before beginning to increase again around age 40. Prior to the advent of modern medicine, most deaths were either before age 5 (infant mortality) or after around age 40 (when diseases of aging begin to appear).

3.1. Hazard ratios and hazard proportionality

A *hazard ratio* is the ratio between two hazard values. For example, we may have two groups of subjects (e.g. people exposed or not exposed to a risk factor), with each group having a hazard function $h_k(t)$ where $k = 0, 1$ corresponds to unexposed and exposed people, respectively. The hazard ratio at time t is $h_1(t)/h_0(t)$. This is a very useful measure of the “instantaneous risk” associated with an exposure. For example, if the hazard ratio is 2 then exposed people have twice the instantaneous risk of experiencing the event as non-exposed people.

Under an assumption of *proportional hazards* hazard ratios are constant in time, meaning in the two-group setting that $h_1 \propto h_0$ for all times t . As we will see below, many popular methods for survival analysis assume proportional hazards, but it is important to note that this assumed proportionality may not always hold in practice.

One approach to exploring possible non-proportional hazards is through the use of *Schoenfeld residuals*. The derivation is somewhat involved so we do not give it here. The *scaled Schoenfeld residuals* r_{ij}^* are of particular interest. There is a scaled Schoenfeld residual for every unit experiencing the event (i.e. when $\delta_i = 1$) and for every covariate $j = 1, \dots, p$. The residual r_{ij}^* captures the deviation from proportional hazards at time $Y_i = T_i$ for the effect of covariate j . If you smooth the r_{ij}^* against the times Y_i , you will get a visual sense of whether the covariate effect is proportional (if the smooth is roughly constant), or if the hazard for that covariate changes over time (the pattern of change is reflected in the smoothed residuals).

3.2. Estimating the marginal hazard function

Since the hazard is a rate per unit time, a primitive estimate of the hazard function at time t_i is

$$\check{h}(t) = \sum_i I(t_i \leq t < t_{i+1}) \cdot \frac{e_i/n_i}{t_{i+1} - t_i}, \quad (16)$$

using the notation defined above. However these estimates fail to concentrate around the true hazards $h(t)$.

The cumulative hazard function is easier to estimate than the hazard function. The most basic non-parametric estimate of the cumulative hazard function under right censoring with no truncation is the [Nelson-Aalen](#)^o estimator

$$\hat{H}(t) = \sum_{i:t_i \leq t} e_i/n_i. \quad (17)$$

Note that

$$\hat{H}(t) = \int_0^t n(s)^{-1} dE(s) \quad (18)$$

where $n(s)$ is the number of units at risk at time s and $E(s)$ is the cumulative number of events up to time s (both are stochastic processes). The integration is with respect to the “counting measure” dE . It is easy to see that the Riemann integral of the step function $\int_0^t \check{h}(s)ds$ is identical to $\hat{H}(t)$.

Since the hazard function h is the derivative of the cumulative hazard function H , it is possible to estimate h by numerically differentiating a smooth estimate of H .

If $\hat{S}_{\text{KM}}$ is the Kaplan-Meier estimate of the survival function and $\hat{H}_{\text{NA}}$ is the Nelson-Aalen estimate of the cumulative hazard function, the identity $S_{\text{KM}} = \exp(-\hat{H}_{\text{NA}})$ fails to hold. This gives rise to another estimator of the marginal survival function called the “Breslow” estimator, defined as $\hat{S}_B = \exp(-\hat{H}_{\text{NA}})$. This estimator of the survival function is more biased than the Kaplan-Meier estimator, but is sometimes used since it preserves the expected relationship between survival probabilities and cumulative hazards.

3.3. Proportional hazards regression

Survival regression is any method that aims to model conditional distributions $P(T|X)$, where T is an event time variable possibly subject to censoring and/or truncation, and X is a vector of explanatory variables. If T is fully observed, specialized techniques for survival regression are not needed. For example, we may regress T or $\log(T)$ on X using least squares or a generalized linear model (GLM). This type of direct approach has been extended to accommodate censoring and truncation, leading to the so-called *transformation models* and *accelerated failure time* (AFT) models that we will not discuss further here.

The most widely-used approach to survival regression is arguably the semi-parametric *proportional hazards* (PH) model, often known as the “Cox model”. This model is explicitly expressed in terms of the hazard function:

$$h(t|X = x) = \exp(\beta'x)h_0(t). \quad (19)$$

The *baseline hazard function* h_0 is unknown and arbitrary, i.e. it is not assumed to follow any parametric family. This is therefore a *semi-parametric* model since it has a finite-dimensional parameter of interest β and an infinite-dimensional nuisance parameter h_0 . It turns out that it is possible to estimate β using partial maximum-likelihood technique without simultaneously estimating h_0 . This exploits the fact that h_0 is orthogonal to β . The cumulative baseline hazard function $H_0(t) = \int_0^t h_0(s)ds$ can be estimated in a separate step if desired, using a modified version of the Breslow estimator discussed above.

When interpreting the results of a PH model, remembering that it is based on proportionality of the hazard function is key. Thus, a given regression slope β_j is the *log hazard ratio* that compares the hazard functions for two individuals who differ by one unit on variable X_j , and have identical values for all other variables. The estimated hazard ratio for the j^{th} covariate is simply $\exp(\hat{\beta}_j)$. Since the PH model assumes proportionality of the hazard functions, this hazard ratio does not depend on t (although the true hazard ratio may depend on t if the PH model is incorrect).

The PH model is essentially a single-index model fit with (partial) maximum likelihood techniques. Thus, once the concepts of the hazard function and proportionality of hazard functions are understood, familiar strategies for regression modeling can be employed. For example, we can include interactions, splines, covariate transformations, and conduct step-wise model searches. There are also versions of information-based criteria such as AIC and BIC for model selection with PH models. The PH model can be extended to accommodate left truncation and competing risks.

As indicated above, proportionality of the conditional hazard functions is a critical assumption in the PH model. This is not always an easy assumption to check, but there are some methods based on residuals that can be employed. Independent censoring is also an important assumption for the PH model to be meaningful.

3.4. Additive hazards regression

Although much less common than the proportional hazards approach, another approach to survival regression based on hazard functions exists which is useful in some particular situations. In the *Aalen additive hazard regression model*, the population hazard function has the form

$$h(t | x) = h_0(t) + \sum_{j=1}^p \alpha_j(t) x_j(t), \quad (20)$$

where the $x_j(t)$ are time-varying covariates, and the $\alpha_j(t)$ are time-varying coefficients. The presence of both forms of time-varying structure makes this model very flexible. The cumulative coefficients $\beta_j(t) = \int_{s=0}^t \alpha_j(s)ds$ play an important role in the estimation and interpretation for this model. If a covariate function $x_j(\cdot)$ increases by one unit from time 0 to time t , then the cumulative hazard of the event being modeled increases by $\beta_j(t)$.

Notably, this model can be fit using ordinary least squares, giving what is essentially a *linear probability model*. For each time t that is an observed event time ($t \in \{t_1, t_2, \dots, t_m\}$), let X^t denote the design matrix based on the covariates at time t , including only subjects who are

at-risk at time t . Similarly, let y^t denote a vector indicating all subjects who have the event at time t , also restricted to subjects at risk. Let $\hat{\alpha}_t$ denote the ordinary least squares slopes for the linear regression of y^t on X^t . Then define $\hat{\beta}_t = \sum_{s \leq t} \hat{\alpha}_s$. These $\hat{\beta}_t$ are estimates of the cumulative coefficient functions $\beta_j(t)$.

The values of $\frac{\hat{\alpha}_{t_j}}{t_{j+1}-t_j}$ estimate the coefficient functions $\alpha_{j(t)}$, but there is very little information available for this estimation – if there are no ties, then only one failure occurs at each time t_j . Therefore, the $\frac{\hat{\alpha}_{j(t_j)}}{t_{j+1}-t_j}$ do not concentrate well around the $\alpha_j(t)$. However the cumulative coefficients behave much better – the $\hat{\beta}_j(t)$ do concentrate around the $\beta_j(t)$. The $\hat{\beta}_j$ are asymptotically normal and confidence bands can be calculated for them, although the procedure for doing so is somewhat complex.

A semiparametric [extension](#)^o of the Aalen additive hazards model allows some of the coefficient functions to be constrained to be constant. Further, it is possible to gain a small amount of additional power by using weighted least squares (WLS) instead of ordinary least squares (OLS). This takes account of the heteroscedasticity inherent in the fact that each y^t has binary elements.

In practice, the additive hazard model is most often used when there are one or more time-varying covariates with frequently changing values, and/or when the hazards are unlikely to be proportional, or when they change over time. Also, the additive hazards model is sometimes used without inference to provide a quick visual check of the shape of the hazard functions and to assess covariate effects.

3.5. Cause-specific hazard regression

If there is a competing risk, it may be meaningful to fit a *cause-specific hazard regression*, treating subjects as being censored when they experience the competing risk. As noted above when discussing the Kaplan-Meier estimator, it is important to be careful when treating a competing risk as a form of censoring. When estimating the marginal survival function, it is rarely advisable to do this, due to the risk of introducing dependent censoring. However in the case of survival regression (proportional or additive), it may be plausible that we have independent censoring after conditioning on covariates.

4. Time-varying covariates

In a basic survival regression analysis, the covariates are defined at “baseline”. That is, every covariate X_j is known at time zero and its value cannot change. There are various approaches to survival regression that can accommodate *time-varying covariates*, e.g. if a subject’s status changes in a way that changes their risk for the event of interest. In doing so, it is important to avoid inadvertently treating a covariate as being defined earlier than it is defined in reality. For example, suppose we are interested in lifespan, and wish to compare lifespan based on occupation. People don’t have an occupation until they are adults, and some occupations (like CEO) are typically determined later in life than others (like athlete). The period between birth and when your occupation is set is a type of *immortal time* – if you became a CEO at age 43 then you were effectively immortal until you reached that age (if you had died before age 43,

you would not have become a CEO). This bias is exaggerated when different occupations tend to have different immortal times.

The Aalen additive hazards model addresses this by allowing the covariates to inform the hazard at the time the covariate is defined, but not into the future. The Cox PH regression model discussed above can be extended to accommodate time-varying covariates, by breaking each person’s observation time into non-overlapping *spells* defined by the times when covariate values change. People are censored at the end of each spell except the final one, where they may (or may not) experience the event of interest.

5. Cumulative incidence functions

Suppose we are in the competing risks setting, where we have a primary endpoint of interest that occurs at time T , and another “competing” event that occurs at time D . A person is censored at time R . The *cumulative incidence* of the primary endpoint is a function $CI(t)$, telling us the probability of having the primary event before time t . It is defined as

$$CI(t) = \int_0^t h_0(t) S_a(t) dt, \quad (21)$$

where $S_{a(t)}$ is the *all cause survival function* and h_0 is the *cause-specific hazard function*. The all cause survival function is the probability of not having experienced either the primary or the competing event by time t . It can be estimated by defining $Y = \min(T, D)$ and using the product limit estimator of the survival function. The cause-specific hazard function is the hazard for having the primary event at time t , given that you are at risk at time t . It can be estimated using the Nelson-Aalen estimator of the hazard function, treating death as a form of censoring.

Cumulative incidence is a convenient way to handle competing risks and avoids the challenges of working with so-called *sub-distribution hazards*, which many people find confusing. It is also possible to introduce covariates into the analysis, to assess how subjects with different characteristics have different cumulative incidences.

Cumulative incidence is most useful when you are interested in the burden (e.g. social or economic) of a condition on individuals or on society. For example, if the primary event is dementia and the competing risk is death, we may want to know the cumulative incidence (probability) of getting dementia by, say, age 85. This probability includes people who get dementia and subsequently die before age 85 (i.e. it is not the proportion of living 85 year old people with dementia). On the other hand, if someone dies young, say at age 60, then they are “saved” from getting dementia by their death. As a result, high risk subpopulations like smokers may appear to have a lower cumulative incidence of dementia, even if smoking increases both the risk of dementia and the risk of death. In this setting, the cumulative incidence function would be telling us that smokers are less likely to experience dementia in their lifetimes than non-smokers.

6. Common biases in survival analysis

The dynamic and partially observed nature of survival data can give rise to many types of bias that are not always obvious. We have already discussed several of these above, such as biases that may result from dependent censoring, non-proportional hazards (when proportionality is assumed), ignoring truncation, and immortal time bias. Like any regression analysis, biases can emerge due to omitted covariates, and in the case of survival analysis, these covariates may affect the outcome times, and/or the censoring times.

7. Parametric and non-parametric methods

As in other areas of statistics, survival analysis can be conducted using parametric or non-parametric methods. Moreover, “semi-parametric” methods play an important role in survival analysis. While parametric methods can be useful, survival analysis tends to emphasize non-parametric and semi-parametric methods over parametric methods.

The most elementary parametric distribution used for survival times is the [exponential distribution](#)[°], although this is usually too simplistic of a model to use in practice. The most commonly-encountered parameterized distribution in survival analysis is arguably the [Weibull distribution](#)[°], and Gamma and log-normal distributions are also encountered.

8. Pseudo-observations

A *pseudo-observation* (or *pseudo-value*) is a synthetic datapoint that combines the observed time Y and censoring status δ into a single real number. The resulting value can then be used in many forms of statistical analysis, for example as an independent or dependent variable in a regression, or in a multivariate analysis such as PCA. Importantly, as discussed in more detail below, the pseudo-observations are approximately independent and their standard deviation reflects the underlying uncertainty in estimating the parameter of interest. Using pseudo-observations allows survival analysis (both estimation and inference) to be conducted using general-purpose statistical methods instead of requiring specialized methods.

Here we will discuss pseudo-observations for the survival probability (i.e. the function $S(t)$ evaluated at a specific time t) and for the Restricted Mean Survival time (RMST, to be defined below). It is also possible to construct pseudo-observations for other quantities such as the cumulative hazard.

Pseudo-observations are closely related to the [jackknife](#)[°]. To motivate the technique, let $\bar{X} = (X_1 + \dots + X_n)/n$ be the sample mean of n observations from a common distribution. Let $\bar{X}_{-i}$ denote the “deleted” version of this statistic (the sample mean with the i^{th} observation deleted). These statistics satisfy the identity $X_i = n\bar{X} - (n-1)\bar{X}_{-i}$. Now consider the more general setting where we have a statistic $\hat{\theta}_n$ based on the full sample of size n , and then we compute this statistic while deleting observation i , to yield $\hat{\theta}_{-i}$. The pseudo-observation is defined to be

$$u_i \equiv n\hat{\theta}_n - (n-1)\hat{\theta}_{-i}, \quad (22)$$

and is interpreted as the contribution of X_i to the statistic of interest, $\hat{\theta}_n$.

It can be shown that the u_i are approximately independent, in that

$$\text{cor}(u_i, u_j) = O(1/n). \quad (23)$$

This leads to an interpretation of pseudo-observations as the “unique contribution of observation i to the estimate”.

The average pseudo-observation is close to the original estimate

$$\text{Avg}(\{u_i\}) \approx \hat{\theta}_n. \quad (24)$$

When the statistic is linear in the data, this approximation becomes an identity, but when the statistic is nonlinear, the value of $\text{Avg}(\{u_i\})$ is a less biased estimate than $\hat{\theta}_n$. Specifically, it has bias of order $O(1/n^2)$ rather than the usual bias of order $O(1/n)$.

We can also obtain a standard error of $\hat{\theta}_n$ from the pseudo-observations, using

$$\frac{\text{SD}(u_i)}{\sqrt{n}} \approx \text{SE}[\hat{\theta}_n]. \quad (25)$$

Thus, the pseudo-observations approximately convert estimation and inference for θ into a linear inference problem, analogous to estimating the population mean with the sample mean. This idea could be useful in many settings, but is particularly useful in the setting of survival analysis since we can now treat the pseudo-observations u_i like any other collection of independent quantitative measurements, and analyze them using a wide variety of statistical methods that are not otherwise adapted to survival analysis.

The most common construction of pseudo-observations for survival analysis is based on the Kaplan-Meier (product limit) estimate of the marginal survival function. We can compute $\hat{S}(t)$ using all data, and then we can compute $\hat{S}_{-i}(t)$ by deleting observation i . There are fast approximations for doing this on large samples without repeating the full calculation for each i . The pseudo-observation is $u_i(t) = n\hat{S}(t) - (n-1)\hat{S}_{-i}(t)$. These can be used, for example, in a regression analysis, regressing u_i on covariates x_i , since it can be shown that $E[u|x]$ can be interpreted as the conditional probability of surviving to time t when the covariates are equal to x .

A common use of pseudo-observations is to perform a regression analysis for the restricted mean survival time (RMST). The RMST is defined as $\int_0^r S(t)dt$. When $r = \infty$ this is just the usual mean. Setting $r < \infty$ is used to reduce the influence of large values. The RMST can be estimated using the Kaplan-Meier estimate of the survival function, by taking $\sum_{i:t_i \leq r} \hat{S}(t_i)(t_{i+1} - t_i)$. Let RMST_i denote the pseudo-observation of RMST for unit i . We then regress the RMST_i on covariates x_i using any regression technique (preferably one that is robust to heteroscedasticity), to understand how the RMST varies with covariates.

There are a few limitations of the pseudo-observation approach, but for each limitation there are work-arounds. Pseudo-observation conditional variances, $\text{var}[u|x]$, are generally not constant in x , i.e. there is heteroscedasticity. Therefore, typically regressions involving pseudo-observations are fit with robust regression techniques such as using the Huber-White type of heteroscedasticity-robust inference. A second issue is that pseudo-observation regres-

sions require unconditional independent censoring (T_i must be independent of R_i without conditioning on the covariate X_i). This limitation can be resolved by using inverse probability weighting to account for censoring that depends on measured covariates, or by calculating the pseudo-observations within strata defined by the informative covariates.